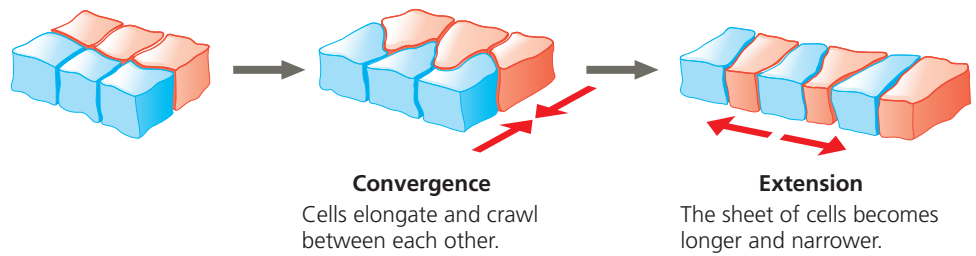


► **Figure 47.17 Convergent extension of a sheet of cells.** In this simplified diagram, the cells elongate coordinately in a particular direction and crawl between each other (convergence) as the sheet becomes longer and narrower (extension).



larval growth, and is then eliminated during metamorphosis into the adult form (see Figure 45.22).

Apoptosis may also occur when a large set of cells is formed, but only a subset has the properties required for further function. Such is the case in both nervous and immune system development. In the vertebrate nervous system, for instance, many more neurons are produced during development than exist in the adult. Neurons that make functional connections with other neurons typically survive; many of the rest undergo apoptosis. Similarly, in the adaptive immune system, self-reactive cells—cells with the potential to attack the animal itself rather than invading pathogens—are often eliminated by apoptosis.

Some cells that undergo apoptosis don't seem to have any function. Why do such cells form? The answer can be found by considering the evolution of amphibians, birds, and mammals. When these groups began to diverge during evolution, the basic developmental program for making a vertebrate body was already in place. The differences in present-day body forms arose through modification of that common developmental program. For example, the shared developmental program generates webbing between the embryonic digits, but in many birds and mammals, including humans, the webbing is eliminated by apoptosis (see Figure 11.21). This is one reason why such different adult forms arise from early vertebrate embryos that look so much alike.

As you have seen, cell behavior and the molecular mechanisms underlying it are crucial to the morphogenesis of the embryo. In the next section, you'll learn some ways in which shared cellular and genetic processes ensure that particular types of cells each end up in the right place.

#### CONCEPT CHECK 47.2

1. In the frog embryo, convergent extension elongates the notochord. Explain how the words *convergent* and *extension* apply to this process.
2. **WHAT IF?** Predict what would happen if, just before neural tube formation, you treated frog embryos with a drug that enters all the cells of the embryo and blocks the function of microfilaments.
3. **MAKE CONNECTIONS** Unlike some other types of developmental abnormalities, neural tube defects are largely preventable. Explain (see Figure 41.4).

For suggested answers, see Appendix A.

#### CONCEPT 47.3

### Cytoplasmic determinants and inductive signals regulate cell fate

During embryonic development, cells arise by division, take up particular locations in the body, and become specialized in structure and function. Where a cell resides, how it appears, and what it does define its development *fate*. Developmental biologists use the terms **determination** to refer to the process by which a cell or group of cells becomes committed to a particular fate and **differentiation** to refer to the resulting specialization in structure and function. You may find it a useful analogy to think about determination being equivalent to declaring a major in college and differentiation being comparable to taking the courses required by your major.

Every diploid cell formed during an animal's development has the same genome. With the exception of certain mature immune cells, the collection of genes present in a given cell is the same throughout the cell's life. How, then, do cells acquire different fates? As discussed in Concept 18.4, particular tissues, and often cells within a tissue, differ from one another by expressing distinct sets of genes from their shared genome.

Even animals that display widely differing body plans share many basic mechanisms of development and often use a common set of regulatory genes. For example, the gene that specifies where eyes form in a vertebrate embryo has a close counterpart with a nearly identical function in the fruit fly *Drosophila melanogaster*. Indeed, when the gene from a mouse is experimentally introduced into a fly embryo, the mouse gene directs eye formation wherever it is expressed in the fly.

A major focus of developmental biology is to uncover the mechanisms that direct the differences in gene expression underlying developmental fates. As one step toward this goal, scientists often seek to trace tissues and cell types back to their origins in the early embryo.